Next-Gen AI Compiler

Agents On The Control Plane

Agents Own Search · Orchestration · Synthesis.
Compilers Own Lowering · Legality · Measure · Fallback.

Ask: Is The Hybrid Prediction (∼2027–31) The Right Bet For Your Roadmap?

Expert Briefing · ∼25–30 min + Q&A · Prediction-first

Agenda

1. Trends · Keep Substrate
2. Verdict + Architecture
3. Roadmap Checkpoints
4. Commercial Blockers + Gaps
5. Technical Prediction (Techniques)
6. Stance · Ask · Appendix

Trend backdrop — two stacks converging

Compilers for AI
mature · fragmented substrate

AI for compilers
rapid hybrid growth

Next-gen ≠ throw away LLVM/MLIR — make them **agent-addressable**.

Six active trends

A — Hybrid guidance
Free IR rewrite fragile
(mlirAgent < identity)
Constrain actions: hints /
advisory metadata / pass lists
AgentCompile · HintPilot · LLM Compiler

B — RL → agents
CompilerGym → tool agents
Heuristics as shippable C++
Workflow compile · freeze · place
Compiler-R1 · Magellan · FlowCompile

C — MLIR + Triton
PyTorch→Inductor→Triton
Also StableHLO→XLA/IREE
Peak often vendor libs / Tile
MLIR shared; product paths diverge

D — Kernel agents
KernelBench: correct *and* faster
One-shot often <20%; fusion hard
GEAK · KernelLLM · AgentCompile
Refine ≠ always faster

E — Verify in the loop
Unit/golden → numerical
→ Alive2-class local formal
Weak on GPU races / FP noise
Stacked oracle ladder for admit

F — Broader object
Mid-decode diagnosis
FMware: prompts / agents / knobs
Agents as compiler engineers
Compiler.next · Magellan · Claude C

Keep substrate, change control plane

Preserve

deterministic lowering
library kernels
build-CI regressability
local formal checks

Adopt

advisory search
multi-agent orchestration
heuristic synthesis
profile / verifier feedback

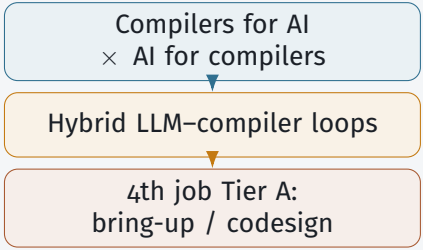
Anti-pattern: replace `opt`/Inductor with unconstrained LLM codegen

Executive verdict

Agents reshape the control plane more than they replace the data plane.

Control plane: search / orchestrate / synthesize

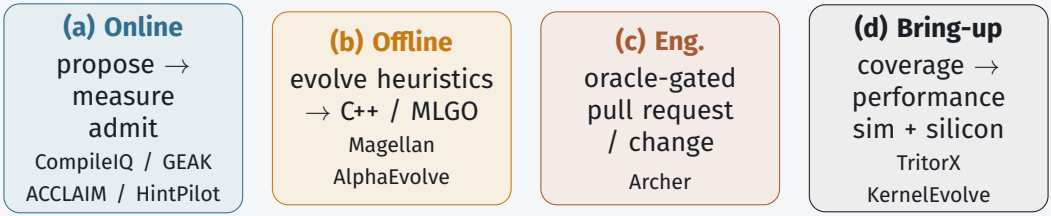
Data plane: lower / legality / measure / fallback



Will not happen soon

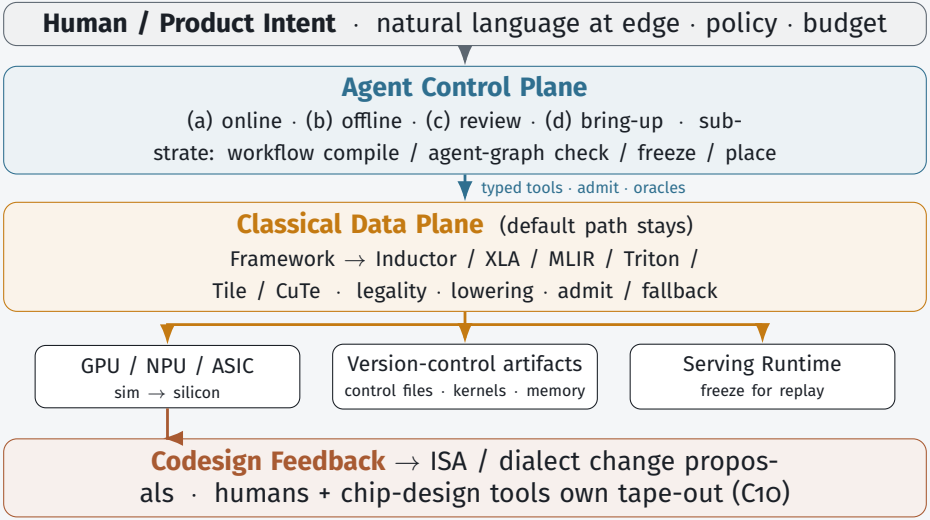
- Unconstrained LLM replaces Inductor/`opt`
- Autonomous chip tape-out via compiler agents (C10)

Four agent jobs — architecture spine



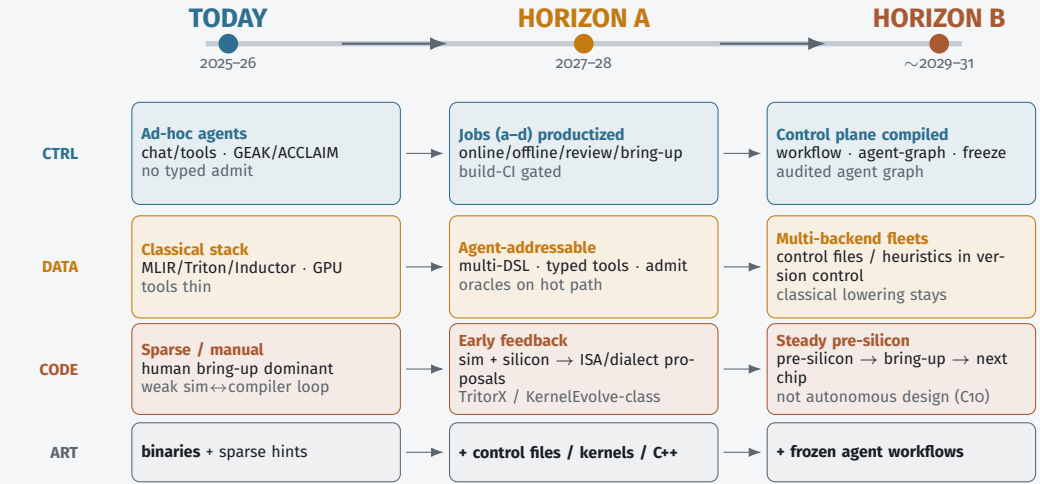
Control plane jobs sit *above* classical lowering — not instead of it.

Architecture — Target Stack



Invariant: LLM guides search — does not silently define unchecked executable behavior.

Architecture evolution — component changes



Components thicken left→right; data plane stays; agent graph becomes compiled & amortized.

What ships / does not by 2028

Predicted to ship

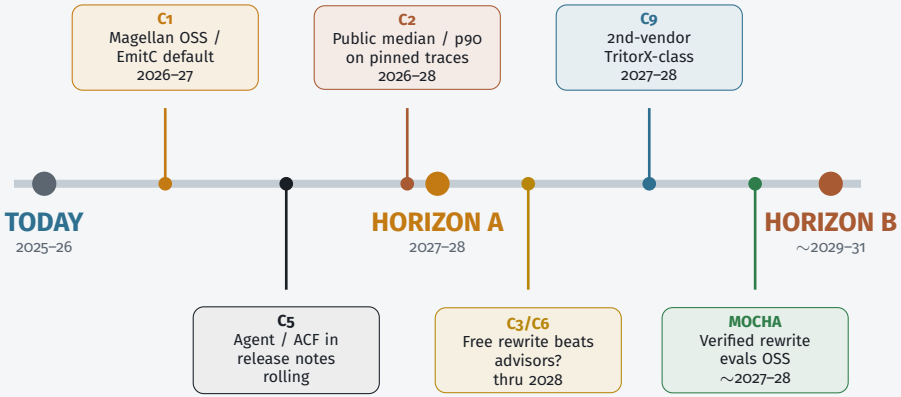
- Agent-addressable tool interfaces
- Hot-path specialize (not silent default-all)
- Magellan *and* MLGO both live
- Oracle pull-request review in serious orgs
- Coverage-first ASIC bring-up
- Triton-family primary; multi-DSL rising

Does *not* ship

- Unconstrained LLM replaces `opt`/Inductor
- One agent IR for all vendors
- Kernel agents uniformly beat eager on fusion ladders
- Autonomous microarch tape-out

Horizon A success = build-CI gated specialize + oracles + freeze artifacts

Roadmap Checkpoints — when the prediction changes



Falsifiable milestones — watch settlement signals; do not average disagreements.

Checkpoint C1 — heuristics vs neural advisors

A — Magellan path

evolve shippable C++ heuristics

Evolve shippable C++ (EVOLVE-BLOCK)

Offline coding agent *is* the control-plane output

Product = evolutionary coding agent + review

B — MLGO / EmitC path

MLGO = learned in-tree LLVM advisors

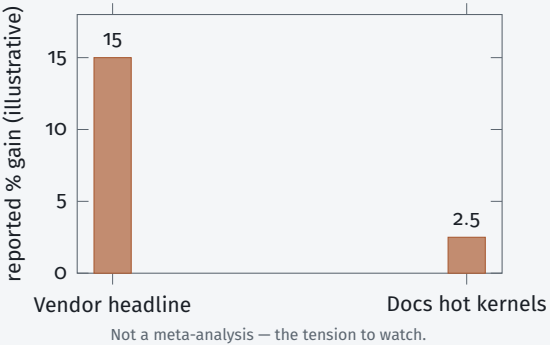
NN advisors stay in-tree

June 2026 plan of record: internal inliner → Android/Fuchsia → Chrome multi-model

Agents train/feature NNs

Settlement: public Magellan llvm patches that displace MLGO *or* EmitC-MLGO becomes customer default
Watch LLVM syncs + OpenEvolve recipes quarterly through 2027

Checkpoints C2 + C5 — gains & default path



Not a meta-analysis — the tension to watch.

C2 "Agents become default" needs **median (p50) build-CI wins**, not best-kernel blogs.

p50 / p90 = 50th / 90th percentile of the gain distribution across builds.

C5 Online specialize (control files / hints) vs offline eng (Magellan/MLGO) — both may live; *which is the default flag?*

Settlement: pinned public traces + release notes listing agent / control-file workflows.

Checkpoints C3 / C6 / C9 / C10 — interface width & scope

C3 — free rewrite vs advisory
Flip: shared suite where free rewrite wins
⇒ wide rewrite interface vs narrow control files / hints

C6 — replace vs control plane
Flip: production default with *no* admit
⇒ agents replaced the compiler (we reject this)

C9 — coverage vs peak
Flip: TritorX → KernelEvolve ladder public
⇒ SKU: coverage service target then performance target

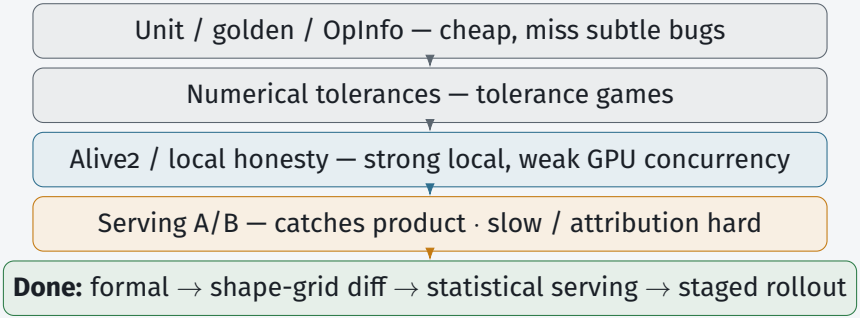
C10 — codesign vs auto tape-out
Flip: microarch primarily agent-proposed *and* compiler-oracle validated ⇒ enters chip-design tooling

Commercial blockers — top 5

1. **Oracles strong enough for money**
   Fast-but-wrong · liability
2. **Cost / replay / when agents run**
   Nondeterministic $N compiles
3. **Agent↔compiler contract**
   Natural-language-only = demo
4. **Distributional production evidence**
   No default-path A/B
5. **Ownership · supply chain · human review**
   Unowned agent code in trusted base

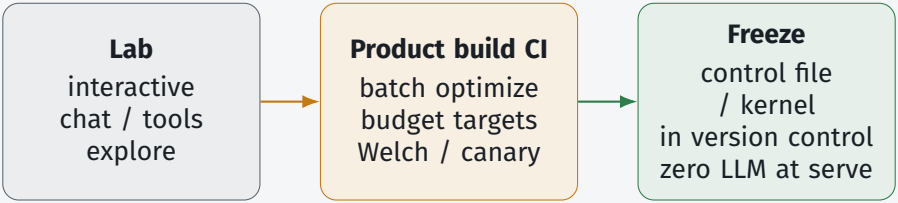
Each blocker gets one slide — productization gaps, not research wishlist.

Blocker 1 — Oracles for money



Room question: who owns false negatives when admit passes and production miscompiles?

Blocker 2 — Cost, replay, when may the agent run?



Cache key: (IR hash, hardware, compiler ver, agent policy) · budget $/%gain

Falsifies commercially: “chat with compiler on every build” without spend/latency targets.

Blocker 3 — Agent↔compiler contract

A. Natural language + pasted logs — demo only

B. Structured admit traces — build CI / bill of materials

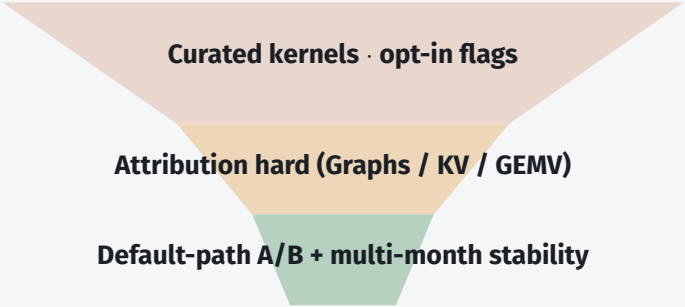
C. Typed tool interfaces — required action space

D. Hybrid — natural-language view over traces

```
admit_record = {graph_hash, hw_id, compiler_ver, action[], oracle[],
artifact_digest, policy_id}
```

Lean: **C + B**; Natural language is a view, not source of truth. Advanced Control File = portable compiler knobs.

Blocker 4 — Distributional production evidence



Marketing bar: median + 90th-percentile gains + cost-to-compile · persistent serving throughput, not single-kernel headlines

Blocker 5 — Ownership, security, human review

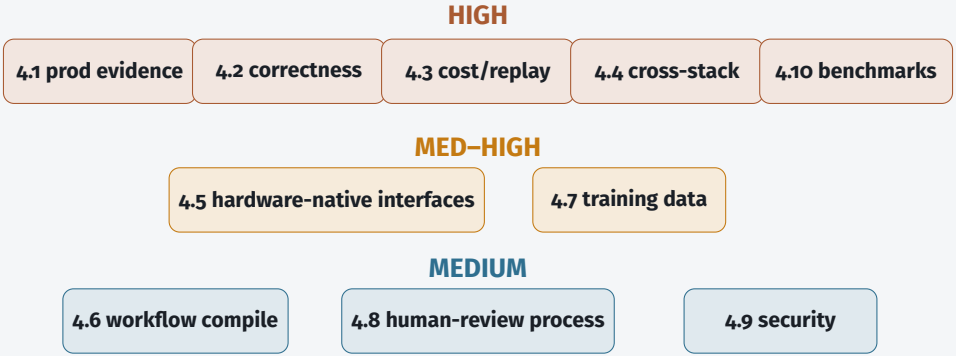
Trusted-base risk
agent kernels / heuristics enter trusted base

Ownership
CODEOWNERS
signed provenance
sandbox

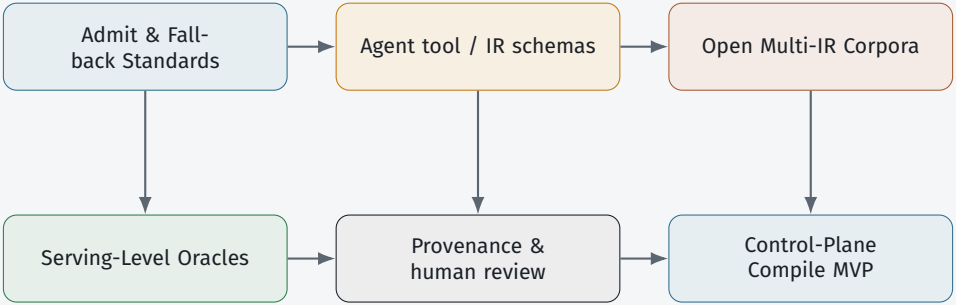
Human-review capacity
agents ↑ drafts
reviewers = bottleneck

Lean: human CODEOWNER + signed admit + sandbox · oracle auto-merge only for **narrow** action classes
Generic forge AI ≠ compiler-oracle review (C7)

Gap map — what blocks the prediction, not a wishlist



Cross-Cutting Research Agenda



Near-term research themes → next: technique map T1–T10 (exists / missing / unlocks).

Technical Prediction — Accelerate The Roadmap

To settle checkpoints and ship Horizon A,
enhance techniques *inside and outside* the compiler (T1–T10).

Within (T1–T5)
T1 Typed agent↔compiler interfaces
T2 Admit / fallback machinery
T3 Control files, hints, fingerprints + replay
T4 Heuristic hooks & in-tree advisors
T5 Dialect / ISA feedback sinks

Outside (T6–T10)
T6 Serving-level oracles & production A/B
T7 Open multi-IR corpora (+ negatives)
T8 Unified benchmark ladder
T9 Provenance, ownership, HITL process
T10 Agent-workflow compile / freeze / place

Enhancing only opt/Inductor/Triton internals is not enough. Next: exists · missing today · checkpoint unlocks.

Technical Prediction — Within The Compiler (T1–T5)

T1 Typed agent↔compiler interfaces Exists: CompileIQ, ACCLAIM/HintPilot, mlirAgent MCP, mlir-opt-repl, FlashInfer Trace. Missing: portable schemas across MLIR·Triton·Tile·StableHLO. Unlocks: C3, C5, C6.

T2 Admit / fallback machinery Exists: AgentCompile checks, Archer oracles, TritonRL verifiers, FlashInfer-Bench gates, VibeServe judges. Missing: shared admit *product* + trusted deterministic fallback. Unlocks: C6 hybrid; money-grade shipping.

T3 Control files, hints, fingerprints + replay Exists: CompileIQ ACFs; FlashInfer Trace + `apply()`. Missing: content-addressed keys (IR × HW × compiler × policy); golden replay on model upgrade. Unlocks: C2, C5.

T4 Heuristic hooks & in-tree advisors Exists: Magellan / AlphaEvolve C++; MLGO advisors; EmitC-MLGO PoR advancing. Missing: settled *default* on named shipping apps. Unlocks: C1; Horizon A job (b).

T5 Dialect / ISA feedback sinks Exists: TritorX / KernelEvolve / Ascend diagnosis (sim+silicon → IR stress). Missing: first-class compiler surfaces for dialect/ISA *change proposals* (humans+EDA still own tape-out). Unlocks: C9, C10.

Technical Prediction — Outside The Compiler (T6–T10)

T6 Serving-level oracles & production A/B Exists: unit/golden/Alive2; **FlashInfer-Bench** serving traces + `apply()`; VibeServe judges. Missing: whole-program/GPU-race/FP contracts; multi-month *default-path* A/B. Unlocks: C2.

T7 Open multi-IR corpora (+ negatives) Exists: Meta LLM Compiler; **KernelBook** → KernelLLM/TritonRL; DRTriton synthetic 100k. Missing: versioned MLIR/Tile/StableHLO + failed/miscompile/slow-correct negatives. Unlocks: selectors beyond LLVM-centric models.

T8 Unified benchmark ladder Exists: KernelBench(-X); FlashInfer-Bench closes a *serving-kernel* rung. Missing: shared IR→kernel→fused→serving ladder with cost-to-compile. Unlocks: honest C2/C9.

T9 Provenance, ownership, HITL process Exists: Magellan reviewable C++; Archer oracle review; sparse SBOM talk. Missing: CODEOWNERS + signed admit records + sandbox for untrusted proposals. Unlocks: trusted-base shipping; C7.

T10 Agent-workflow compile / freeze / place Exists: FlowCompile, Auto, AgentFlow, hetero place; VibeServe (early evidence *now*). Missing: shared agent-graph IR + fail-closed CI (productization → Horizon B). Unlocks: Horizon B control-plane compile.

Technical Prediction — What Each Checkpoint Needs

C1 ← **T4** Magellan-class synthesis *or* EmitC-MLGO customer default on named apps.

C2 ← **T3+T6+T8** replayable artifacts + serving oracles + ladder; FlashInfer-Bench = pressure, not settlement.

C5 ← **T1+T3** typed interfaces + freeze-before-serve named in release notes.

C3 / C6 ← **T1+T2** constrained tools + admit/fallback (lean: hybrid advisory until free rewrite proves out).

C9 ← **T5+T8** coverage→perf agents + dialect sinks; need second-vendor TritorX-class path.

C10 ← **T5** failure modes → ISA/dialect *proposals only*; humans + chip-design tools own tape-out.

Technical Prediction — Critical Missing Parts Now

Highest-leverage bets to accelerate Horizon A

1. Money-grade oracle stack (T2+T6)
local formal → shape-grid diff
→ serving statistics → staged rollout
Without this, agents stay demos

2. Replayable artifact contract (T3)
control files / kernels / heuristics
with cache keys + golden replay
Without this, CI rejects the loop

3. Portable agent compile interface (T1)
summaries · actions · admit records
across MLIR / Triton / Tile / StableHLO
Without this, every stack re-glues

4. Open ladder + multi-IR data (T7+T8)
correctness × speed × $/compile
plus negative (failed) examples
Without this, claims stay incomparable

Survey lean: T1–T5 must ship as product surfaces; T6–T10 are equally first-class — mostly outside classical lowering.

Org Adoption Questions

1. Online (CompileIQ) vs Offline (Magellan) — which matches your release model?

2. Oracle stack: Alive2 / golden / serving A/B — who owns false negatives?

3. Agent contract surface: LLVM / MLIR / Triton / StableHLO / Tile?

4. How do you cache/regress traces across compiler *and* model upgrades?

5. Max $/build for a median X% win? Named maintainer per admitted artifact?

Working stance until conflicts settle

1. Hybrid: agents search; compilers lower (not replace)
2. Magellan || MLGO — parallel bets (C1)
3. Discount single-number speedups (C2)
4. Constrained actions + strong oracles (C3)
5. Demote generic SCM AI (C7)
6. Coverage→performance ladder; no autonomous chip design (C9/C10)

Commercial checklist — handout

Architecture

typed tools + admit traces
version control ≫ dense memory ≫ scratchpad
state-machine plan for service targets
freeze before default-on

Trust

layered oracles
CODEOWNERS + provenance
joint version pins
A/B before “prod default”

Business

sell regressable artifacts + build-CI quota
customer owns outputs
coverage then performance service target
token budget per %gain

Discussion

Thank you — hybrid bet stays the ask

1. Which checkpoint flips your investment first — C1, C2, or C9?
2. Harder commercial bar: **oracle strength** or **replay/freeze**?
3. Building job (a), (b), or (d) first — what do you refuse online?

Ask again: is the hybrid bet right for your roadmap? Watch settlement signals — do not average disagreements.

Appendix

Evidence Store Snapshot

Publications digests · commercial product signals · open repositories.
Tiered for the agentic-compiler prediction — not a full catalog.

Appendix — evidence map

Publications

∼115 digests
one file per source

papers · blogs · talks
forge pages · minutes

★ = prediction-critical

Products

Commercial SKUs as *prediction signals*

Tier A shapes the future
Tier B = data-plane defaults

not a full SKU list

Repositories

GitHub / Gerrit / forge artifacts tiered A/B/C

A = agents reshape compile
C = generic forge AI only

not an exhaustive forge map

Use ★ digests + Tier A products/repos when a checkpoint or blocker is contested.

Appendix — Tier A commercial signals

Company offerings that shape what ships — roles: agent loop / kernel surface / cloud agent

Google / DeepMind — AlphaEvolve Cloud (GA); Magellan + MLGO (offline heuristics *and* neural advisors)

NVIDIA — CompileIQ (+ agent-skills); CUDA Tile / Tile IR; TensorRT-LLM agent skills

AMD — GEAK (multi-agent Triton / Instinct kernel optimization)

Meta — LLM Compiler / KernelLLM; TritorX + KernelEvolve (ASIC bring-up); Helion

FlashInfer / NVIDIA·UW·CMU — FlashInfer-Bench (serving-trace ladder + `apply()` into SGLang/vLLM)

Tier B baselines: TensorRT-LLM · Inductor · XLA/StableHLO · FlashInfer kernels · Modular MAX · OpenVINO · Neuron · Hexagon-MLIR

Appendix — Tier A open repositories

Offline / review / heuristics
OpenEvolve (Magellan-class loop)
HeuriGym (heuristic bench)
Archer (oracle pull-request review)
Compiler-R1 · HintPilot
mlirAgent (negative free-rewrite)

Online / kernels / bring-up
ACCLAIM · CompileIQ
GEAK · KernelAgent
KernelBench / KernelBench-X
FlashInfer-Bench · AutoKernel
Helion · TritorX / KernelEvolve

How to read tiers: A = agents change heuristics/kernels/knobs/review with domain oracles. B = data-plane hosts. C = generic forge AI (demoted for prediction).

Appendix — prediction-critical digests (★)

Highest-signal sources for the hybrid prediction — mechanisms, not headlines

Vision / funded: Compiler 2.0 Ken Kennedy plenary (MIT) · DARPA MOCHA / Aarno Compiler 2.0

Offline / online jobs: Magellan (+ LLVM slides) · ACCLAIM (+ GitHub) · EmitC-MLGO June 2026 plan-of-record minutes

Bring-up / codesign: TritorX · KernelEvolve · Ascend hierarchical diagnosis · KForge

Kernels / ladder / data: CompileIQ agent-skills · FlashInfer-Bench ★ · KernelBook ★ · AutoKernel · CuTeGen · Auto · Flow-Compile

Full digest index: one file per source under the publications store (∼115 entries). Technique map: SURVEY §5.8 T1–T10.

Appendix — publication groups

17 GPU kernels & inference
13 Agentic & RL compilers
13 Source control & review

10 Classic DL compilers
10 Company infra
9 Forums & workshops

8 Surveys & vision
8 Foundation LLMs
6 MLGO & RL gyms

5 HW codesign & bring-up
4 Commercial products
4 Control-plane substrate

How digests are used
Mechanism first; demote generic forge AI.
When sources disagree, keep both sides — do not average.
Update prediction only when Tier A / ★ evidence moves.